

Chapter 13: Sound

Textbook Exercise Solutions & Answers

1. Fill in the blanks with the proper words.

(a) The propagation of sound does not occur through a

vacuum

(b) Noise pollution is a issue.

social/serious

(c) The sound which is disagreeable to our ears is called

noise

(d) Noise has adverse effects on our

health

2. What should we do?

(a) The silencer of a motorcycle is broken.

1. If the silencer of a motorcycle is broken, vehicle should be immediately taken to the garage for repair.
2. If silencer is not maintained properly, it will continue making loud noise on the road while in motion, resulting in increasing noise pollution.

(b) A factory in the surrounding is producing continuous loud noise.

Factories should be located at a proper distance away from residential areas. Authorities should be contacted who will control the decibel level.

3. Write the answers in your own words.

(a) What is meant by vibration?

Vibration of an object is necessary for the production of sound. As long as the object vibrates, the sound is heard. When the vibration stops, the sound also stops.

(b) Explain with the help of practical examples how sound is propagated through solids.

1. Singing
2. Chirping of birds
3. Bursting of crackers
4. Whispering
5. Whistling
6. Flute
7. Honking
8. Dog barking

(c) What is meant by noise pollution?

Continuous noise which has ill effects is called noise pollution.

(d) What measures will you take to control noise pollution?

Measures for preventing noise pollution:

1. As far as possible, we should avoid blowing horns.
2. The volume of the TV or radio in the house should be restricted to those watching the programmes.
3. Vehicles should be maintained properly to reduce the unnecessary sounds they produce.
4. Factories, airports, railway stations and bus stands should be located at the proper distance away from residential area.
5. Planting of trees helps reduce noise as trees act as a natural barrier.

4. Complete the table.

Nature of sound	Unpleasant	Pleasant
1. Speaking		✓
2. Whispering		✓
3. Aeroplane Sounds	✓	
4. Horns of Vehicles	✓	
5. Railway Engine	✓	
6. Rustling of leaves		✓
7. Neighing of a horse	✓	
8. Ticking of a clock		✓
